



- (i) With reference to Fig. 3.2, describe the effect of the different diets on the percentage of *E. coli* that are *lac*⁺.

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..... [2]

- (ii) The scientists concluded that *lac*⁺ *E. coli* had a selective advantage when colonising the gut, but only in the presence of lactose.

Suggest why the presence of lactose caused *lac*⁺ *E. coli* to have a selective advantage.

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..... [2]

[Total: 12]

